Algebra, Geometry, Combinatorics, Number Theory Pre-Test

The Gauss Group

thegaussgroup@gmail.com

1 Introduction

Below is a diagnostic test to determine if students are ready for Algebra, Geometry, Combinatorics, and Number Theory courses offered by The Gauss Group. We suggest using the following process to assess whether or not your student is ready for the courses.

1. Take the following test and try to answer as many questions as possible in 40 minutes.
2. If the score is 5/10 or higher, then it is beneficial to take these courses. If not, the student may need to work on the prerequisites.
3. The answers to the problems are listed on the last page of this document.
4. If you have any questions about the problems below, or find a mistake, please email us.

2 Problems

1. Find a number that when subtracted by 3 and then multiplied by 6 is equal to 36.
2. The center of a circle O is defined at $O(4, -2)$. If a point on the circle is defined at $X(4, 4)$, find the area of circle A in terms of π.
3. Find the solutions to the system:

$$12x + 6y = 19$$

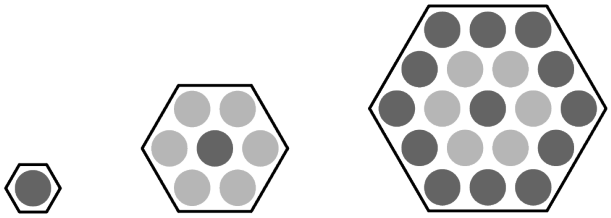
$$2x + 5y = 12$$

4. Find the **sum of the roots** of

$$x^2 + 4x - 2021 = 0$$

5. **(2020 AMC 8)** Four friends do yard work for their neighbors over the weekend, earning \$15, \$20, \$25, \$40, respectively. They decide to split their earnings equally among themselves. In total how much will the friend who earned \$40 give to the others?

6. **(2020 AMC 8)** Three hexagons of increasing size are shown below. Suppose the dot pattern continues so that each successive hexagon contains one more band of dots. How many dots are in the next hexagon?



7. **(2020 AMC 8)** How many positive integer factors of 2020 have more than 3 factors? (As an example, 12 has 6 factors, namely $1, 2, 3, 4, 6$, and 12.)

8. **(2016 AMC 8)** Which of the following numbers is not a perfect square?

$$1^{2016}, 2^{2017}, 3^{2018}, 4^{2019}, 5^{2020}$$

9. **(2016 AMC 8)** Find the value of the expression

$$100 - 98 + 96 - 94 + 92 - 90 + \cdots + 8 - 6 + 4 - 2$$

10. **(2019 AMC 10A)** What is the least possible value of

$$(x + 1)(x + 2)(x + 3)(x + 4) + 2019$$

where x is a real number?

3 Answers

1. 9
2. 36π
3. $x = 23/48$; $y = 53/24$
4. $x = -4$
5. \$15
6. 37
7. 7
8. 2^{2017}
9. 50
10. 2018